NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

Lecture #141 Immunogenetics, Lymphocyte Activation, Immune Regulation and Tolerance Part 1

Bradley Hickey, MD, PhD, MPH, MBA, MA, FACOEM
Assistant Professor, Department of Clinical Medicine
NYIT COM at Arkansas State University
Email: bhicke03@nyit.edu

Office of Academic Affairs

Why These Rules of Immunity Matter

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- These rules evolved to prevent inappropriate or harmful immune activation
- Adaptive immunity follows non-negotiable rules
- These rules determine what the immune system can and cannot respond to
- Understanding the problem explains the mechanism

Learning Objectives / Conceptual Focus

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

- Explain why T-Cells require antigen presentation
- Describe the role of MHC molecules in immune safety
- Connect antigen presentation to tolerance and disease
  - Emphasis on mechanisms rather than memorization

The Unique Role of T Cells

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

- Specialized for detecting intracellular threats
- Focus on viral infection and malignant transformation
- Function as inspectors rather than direct antigen binders
  - Requires intermediary systems to safely interpret antigen information

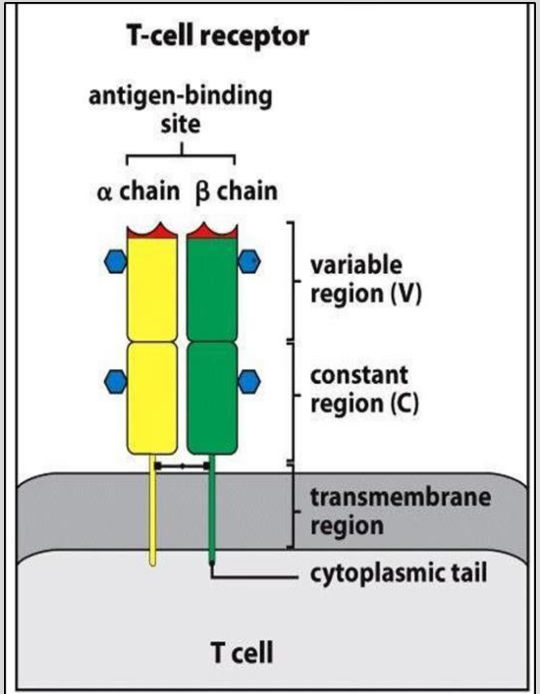


Figure 5.1 The Immune System, 3ed. (© Garland Scienc

T-Cells Do Not Bind Free Antigen

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

- Free antigen recognition would be unsafe
- Processing inserts contextual control
- Prevents responses to harmless exposure
    - Prevents activation based solely on molecular presence

The Major Histocompatibility Complex (MHC)

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

- Forces T-Cell recognition to occur in a controlled cellular context
- Encodes the immune system's peptide display platform
- Known as HLA molecules in humans
- Enables standardized antigen reporting

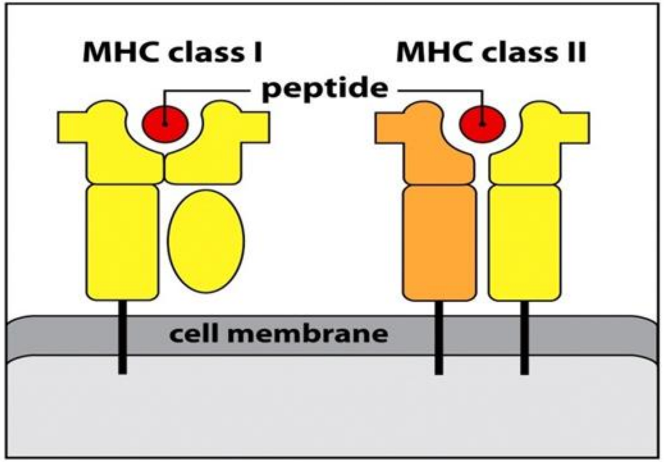


Figure 3.8 The Immune System, 3ed. (© Garland Science 2009)

MHC Polymorphism and Population Survival

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- MHC genes are highly polymorphic
  - No single pathogen can evade all MHC variants simultaneously
- Limits pathogen escape at the population level
- Creates individual variability in immune responses

Clinical Consequences of MHC Diversity

- Differences in disease susceptibility
  - Reflects differences in peptide presentation efficiency
- Variable responses to infection and vaccines
- Central importance in transplantation

MHC Restriction Defines T-Cell Recognition

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

- T cells recognize peptide plus MHC
- Specificity depends on MHC context
- Explains personalized T-cell responses
    - Same antigen may elicit different responses across individuals

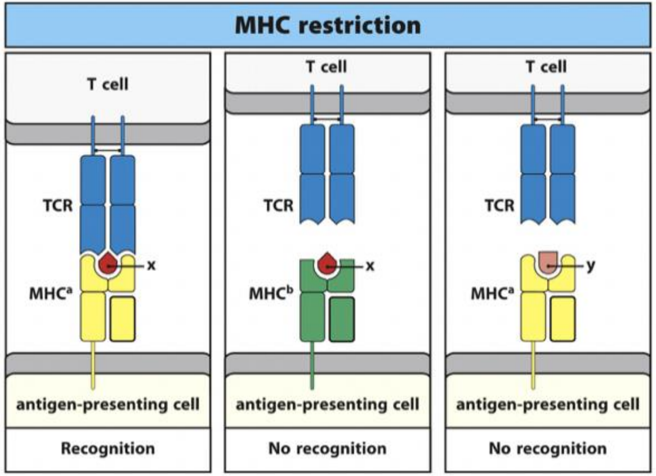


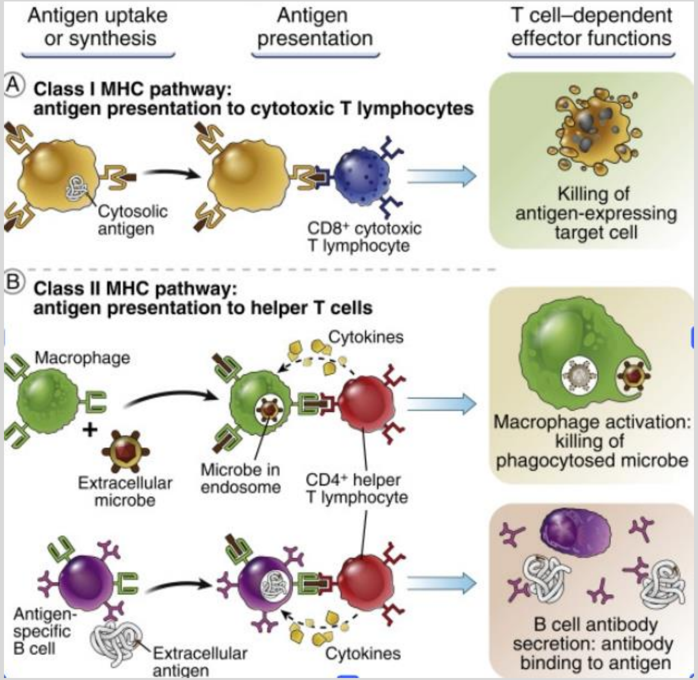
Figure 6.21 Janeway's Immunobiology, 8ed. (© Garland Science 2012)

Why Two MHC Classes Exist

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

- Antigens arise from different cellular locations
- Intracellular vs extracellular threats
- Distinct immune responses required
    - Different threats demand different decision pathways

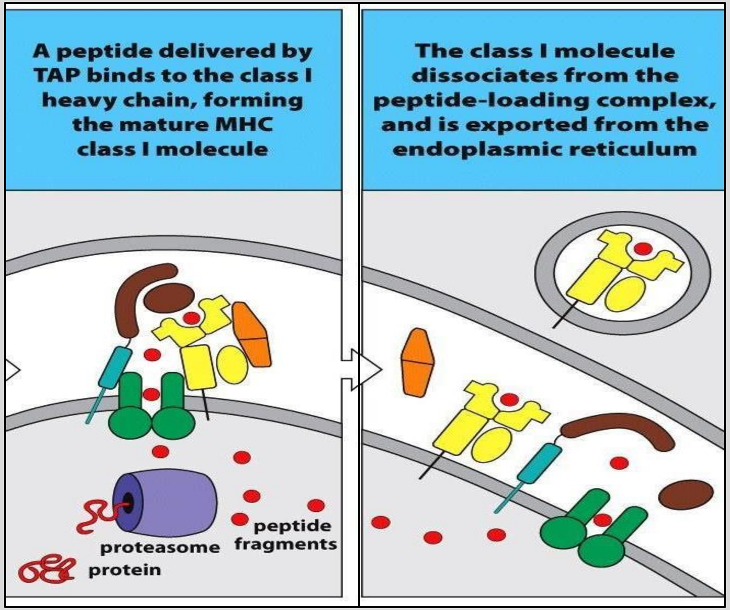


Zanza, et al. Cellular Immuno-Profile in Septic Human Host: A scoping Review. Biology 2022, 11, 1626.

MHC Class I: Continuous Intracellular Surveillance

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Presents peptides from intracellular proteins
- Expressed on all nucleated cells
- Enables early detection of abnormal cells



The Immune System, 3rd Ed., © Garland Science, 2009

Where MHC Class I Peptides Come From

- Generated through normal protein turnover
- Viral and self proteins treated identically
- Antigen sampling is passive, not inflammatory

Proteasomal Processing of Intracellular Proteins

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Proteasomes generate peptide fragments
  - Immune decision making does not occur during protein degradation
- No danger assessment at this stage
  - Antigen generation is decoupled from immune decision-making
- Surveillance layered onto normal metabolism

TAP-Mediated Peptide Transport

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Peptides transported into the ER
- Selective for appropriate peptide size
- Critical quality-control checkpoint

Assembly of MHC Class I Molecules

- Heavy chain associates with β2-macroglobulin
- Peptide binding stabilizes the complex
- Empty MHC molecules are not displayed
  - Prevents surface expression of unstable or misleading complexes
  - Ensures only stable peptide MHC complexes reach the cell surface

CD8 T Cells and Targeted Cell Elimination

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Recognize peptide–MHC class I complexes
- Assess intracellular health of individual cells
- Initiate cytotoxic responses when needed

Continuous Nature of Class I Presentation

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Antigen reporting is non-optional
  - Absence of presentation itself can signal abnormality
- Cells cannot hide from surveillance
- Supports early, localized immune responses

Why a Second Presentation Pathway Is Needed

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

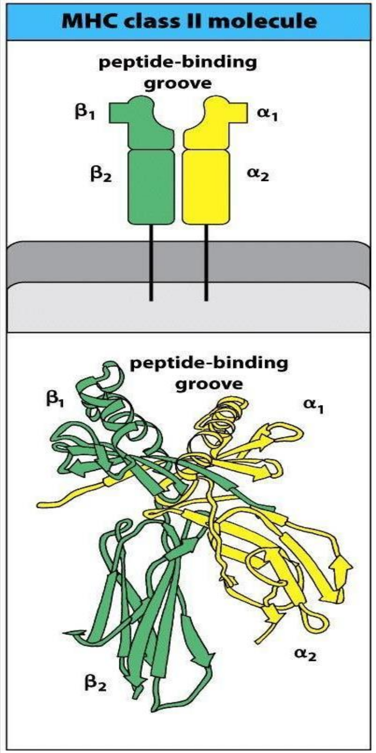
- Some immune decisions must be made at the systems level
  - Individual cell killing is insufficient for coordinated immune responses
- Extracellular threats require coordination
  - Centralizes immune activation authority
  - Antibody production and immune orchestration require helper T-Cell input
- Cytotoxicity alone is insufficient
- Leads to MHC class II pathway

Restriction of MHC Class II Expression

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

- Limited to professional APCs
- Ensures controlled immune activation
- Prevents widespread inappropriate responses



The Immune System, 3rd Ed., © Garland Science, 2009

Uptake of Extracellular Antigens

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

- Antigens internalized by endocytosis or phagocytosis
- Degraded in vesicular compartments
- Maintains separation from class I pathway
  - Prevents extracellular material from entering the Class I surveillance system
  - Prevents extracellular material from triggering cytotoxic surveillance

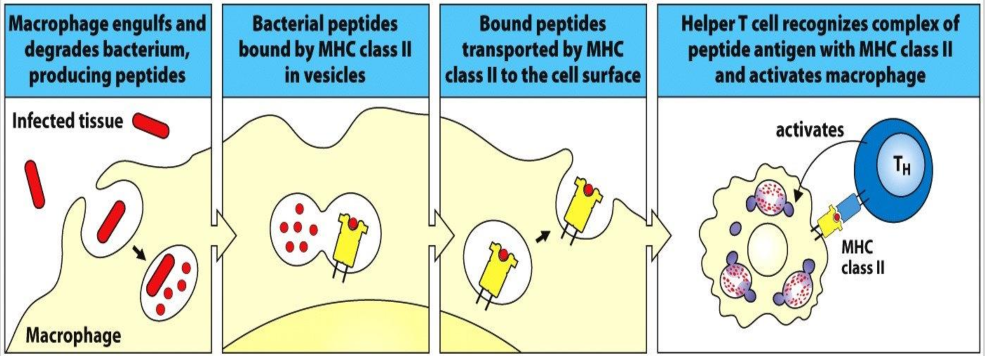


Figure 3.11 The Immune System, 3ed. (© Garland Science 2009)

The Invariant Chain Protects MHC Class II

- Blocks premature peptide binding
- Prevents intracellular peptide loading
- Preserves pathway fidelity

Targeting Class II Molecules to Endosomes

- Invariant chain directs trafficking
- Ensures exposure to extracellular peptides only
  - Spatial routing enforces correct antigen source
- Critical for specificity

Controlled Degradation of the Invariant Chain

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Removed stepwise, not all at once
- Maintains control during transport
- Prevents accidental peptide loading

CLIP as a Temporary Placeholder

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

- Occupies the peptide-binding groove
- Prevents low-affinity peptide binding
- Maintains accuracy over speed

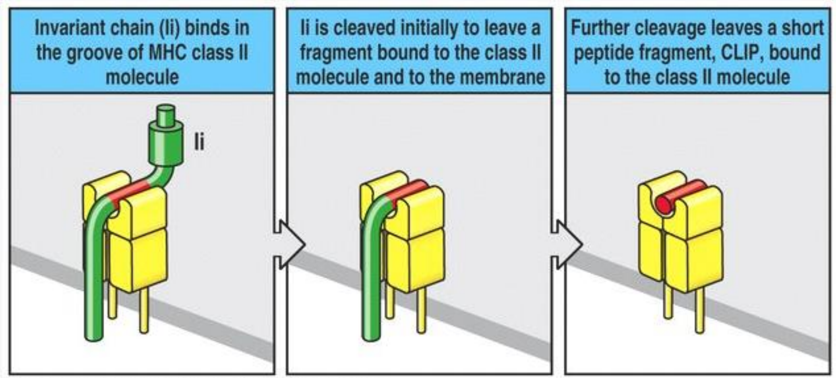


Figure 5-8 part 2 of 2 Immunobiology, 6/e. (© Garland Science 2005)

HLA-DM and Peptide Editing

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Removes CLIP from class II molecules
- Promotes stable peptide binding
- Increases signal-to-noise ratio
    - Favors peptides capable of sustained T-Cell engagement

Surface Display of MHC Class II Complexes

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Displayed only on professional APCs
- Engage CD4 T cells
- Provide instructional immune signals

CD4 T Cells as Immune Coordinators

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Do not directly kill target cells
- Shape immune responses via cytokines
- Influence multiple effector pathways

Why Coordination Matters

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

- Prevents inefficient or damaging responses
- Tailors immunity to threat type
- Reduces unnecessary tissue damage

Revisiting Class I vs Class II Pathways

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

- Intracellular vs extracellular surveillance
- CD8 vs CD4 T-cell engagement
- Distinct but complementary roles
  - Both pathways are required for complete immune decision-making

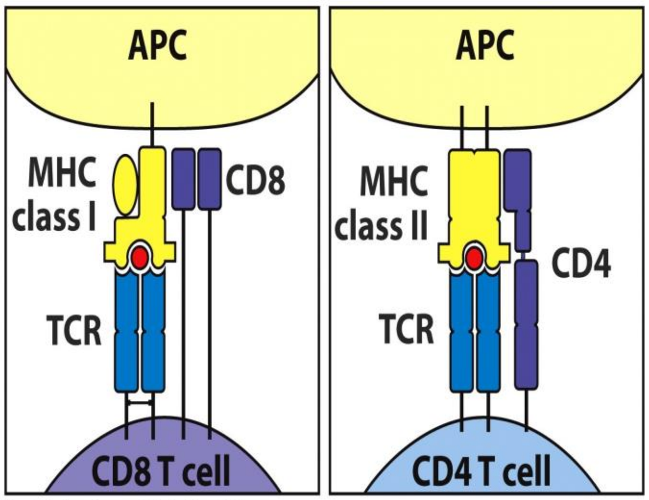


Figure 3.9 The Immune System, 3ed. (© Garland Science 2009)

Antigen Presentation Is Not Enough

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Recognition alone does not activate T cells
  - Antigen recognition is separated from immune action by design
- Prevents responses to harmless antigens
- Separates recognition from action
  - Prevents immune responses to harmless or self antigens

Costimulation:
The Second Signal

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Provides contextual information
  - Confirms that antigen recognition occurs in infection or tissue damage
- Indicates danger or inflammation
- Required for full T-cell activation
  - Absence of costimulation leads to tolerance, not immunity

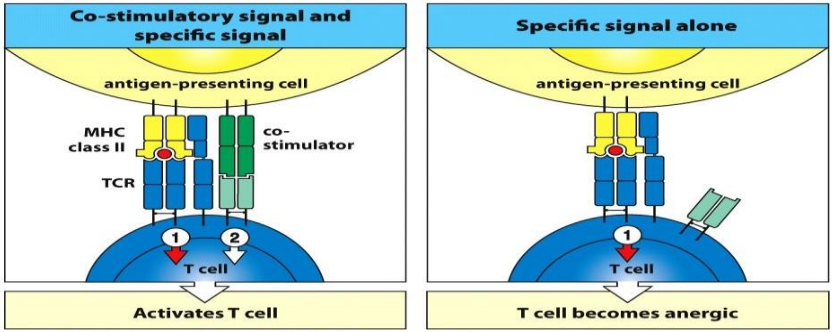


Figure 8.18 The Immune System, 3ed. (© Garland Science 2009)

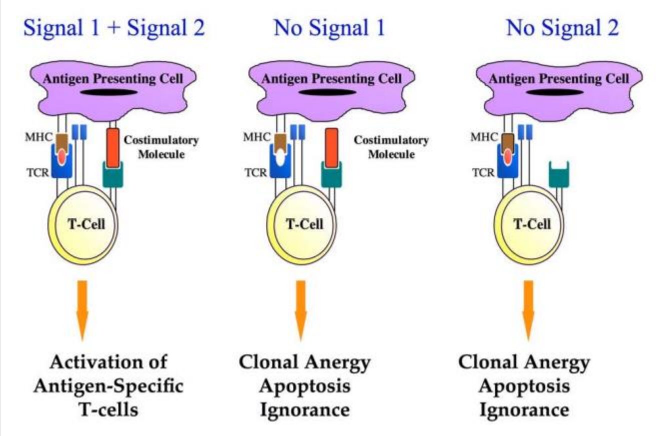
Why Only Professional APCs Activate Naïve T Cells

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Deliver both antigen and costimulation
- Centralizes immune decision-making
  - Prevents widespread, unregulated T-Cell activation
- Supports peripheral tolerance

Antigen Recognition Without Costimulation

- Leads to functional unresponsiveness
- Known as anergy
- Protects against autoimmunity
  - Functionally silences potentially self-reactive T-Cells



Hodge, J. W. et al. Vector-based delivery of tumor-associated antigens and Tcell co-stimulatory molecules in the induction of immune responses and anti-tumor immunity. Cancer Detect Prev 26: 275-291 (2002)

Context Shapes Immune Responses

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Determines whether and how responses occur
- Influences magnitude and type of immunity
- Enables adaptive flexibility

Costimulation and Tolerance Are Linked

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Central tolerance is incomplete
- Peripheral mechanisms provide backup
- Costimulation requirements enforce restraint
  - T-Cells encountering antigen without signal 2 become functionally inactive
  - Failure to deliver signal 2 actively promotes tolerance
  - Peripheral tolerance is an intentional outcome, not an accident

Tolerance Is an Active Process

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Requires continuous regulation
- Multiple checkpoints involved
- Failure leads to disease

Balancing Immune Responsiveness and Safety

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Over-response causes tissue damage
- Under-response permits disease
- Regulation maintains equilibrium

Clinical Relevance

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

- Costimulatory pathways are important targets for clinical intervention
- Therapeutic targets exploit normal immune regulatory checkpoints
  - Checkpoint inhibitors modify costimulatory balance to enhance or suppress immunity
- Therapeutic manipulation can enhance or suppress immune responses

Integrating Antigen Presentation, Costimulation, and Tolerance

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Antigen presentation provides specificity
- Costimulation determines whether a response should occur
- Tolerance mechanisms prevent harmful self-reactivity

Superantigens:

A Major Exception

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

- Bypass normal antigen specificity
  - Violates normal rules of antigen specificity and safety
- Bind MHC and TCR externally
- Trigger widespread activation

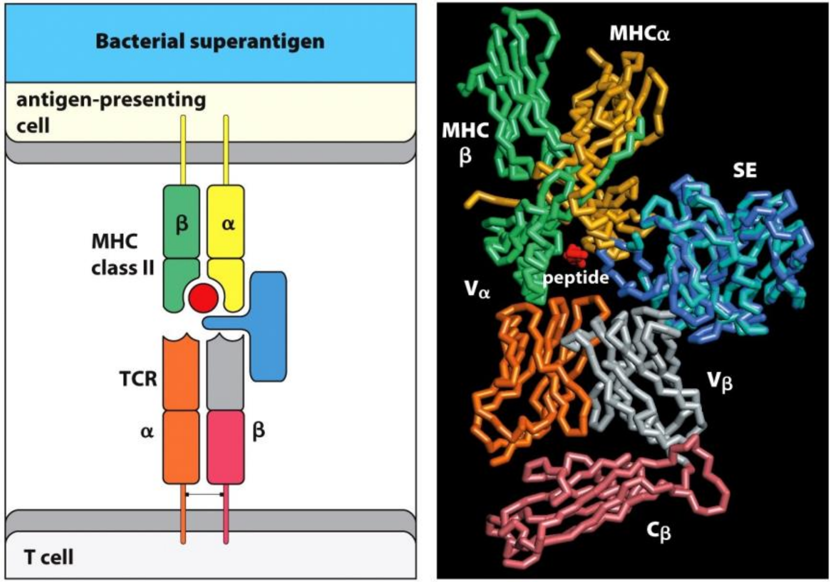


Figure 11.8 The Immune System, 3ed. (© Garland Science 2009)

Consequences of Superantigen Activation

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Massive cytokine release
- Systemic inflammation
- Severe clinical syndromes

Typical Response	Superantigen Response
Antigen must be processed by APC	APC Processing not required
Antigen recognition and T-Cell activation is MHC-II restricted	MHC-II cells required for activation by not MHC-II restricted
Small amount of T-Cells activated	20-30% T-Cell activated

What Superantigens Teach Us About Design

- Normal rules exist for safety
  - Restrictions in immune recognition are intentional, not inefficient
- Shortcuts reveal system vulnerabilities
- Precision is essential

Clinical Examples of Superantigen Exposure

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Bacterial Toxins
- Severe Clinical Syndromes
- Rapid Onset and Severity

MHC Expression Patterns and Immune Outcomes

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- MHC Class I on all nucleated cells
  - Widespread intracellular surveillance
- MHC Class II restricted to APCs

Genetic Variation and Disease Susceptibility

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- MHC alleles influence risk
- Affect peptide presentation efficiency
- Link genetics to immunopathology

Antigen Presentation in Transplantation

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Antigen Presentation is central to transplant rejection
- Recipient T Cells may recognize donor MHC molecules as foreign
- MHC matching is critical

Integrating Antigen Presentation and Regulation

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Presentation provides specificity
- Costimulation provides context
- Tolerance provides restraint

Key Takeaways

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

- Antigen presentation determines immune visibility
- MHC constrains T-cell recognition
- Regulation ensures immune precision

Transition to Downstream Immunology Topics

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Antigen presentation underpins all adaptive immunity
- Concepts recur in hypersensitivity and autoimmunity
- Foundation for future clinical reasoning

Summary Slide

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

- **Summary Point 1**

- Antigen presentation determines **what the immune system can see**
- MHC molecules constrain T-cell recognition to specific, regulated contexts
- This design prevents indiscriminate or dangerous immune activation

- **Summary Point 2**

- Costimulation and tolerance mechanisms determine **whether a response occurs**
- Separation of recognition from action protects the host
- Failures in these controls explain autoimmunity, transplant rejection, and immune evasion

- Core References:

- Abbas AK, Lichtman AH, Pillai S. *Cellular and Molecular Immunology*. Elsevier.
- Murphy K, Weaver C. *Janeway's Immunobiology*. Garland Science.
- Parham P. *The Immune System*. Garland Science.
- Klein J, Sato A. "The HLA system." *New England Journal of Medicine*.

Lecture Feedback Form:

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R